key points

• Quantum entities behave as both particle and wave
• There is an inherent uncertainty built into the quantum world
• We don't notice quantum effects in everyday life because Planck's constant is so small

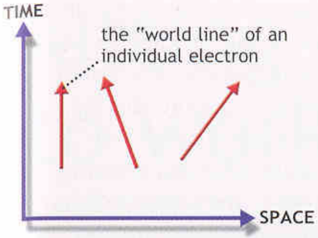
The roots of QED were established in the 1930s, when two physicists, the German Hans Bethe and the Italian Enrico Fermi, suggested that interactions between charged particles could be described in terms of photons (single units of light) being exchanged between the particles. The complete version of the theory emerged in the 1940s from the work of three physicists, Sin-itiro Tomonaga in Japan, and Julian Schwinger and Richard Feynman in the US. Each of the three versions that they developed is mathematically equivalent to each of the others, but Feynman's approach is easier to understand in physical terms, and has become the standard version of QED.

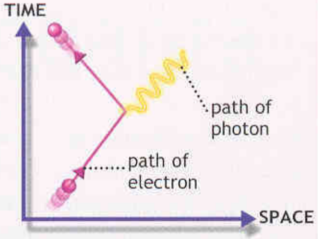
Feynman diagrams

The physics of QED can be understood in terms of pictures called Feynman diagrams. These represent the paths of electrons using lines, known as "world lines" that are placed within two coordinates of time and space. A vertical line indicates a stationary electron because its position in space is not changing. The more angled the world line, the greater its rate of change of position in space and therefore the faster it is moving. A photon associated with the field of a magnet can be introduced into a Feynman diagram. The photon can be shown interacting with an electron traveling in a straight line, and deflecting it onto a new path. This simple version of QED predicts that a property known as the magnetic moment of the electron has a value of exactly one. It doesn't matter just what the magnetic moment is (you can think of it as a measure of how easily an electron can be twisted by a magnetic field), but what does matter is that it is possible to measure this property very accurately in experiments. The experiments show that the magnetic moment is actually a little bit more than one, so this simple version of QED cannot be the final word on the subject.

Feynman showed how the simple model could be improved, and drew a picture to highlight what was going on. In the next step of the calculation, while it is interacting with the photon, the electron also emits and then reabsorbs another photon, interacting with itself. It is easy to draw the picture, harder to carry out the mathematical calculations. When the calculations for this interaction were carried

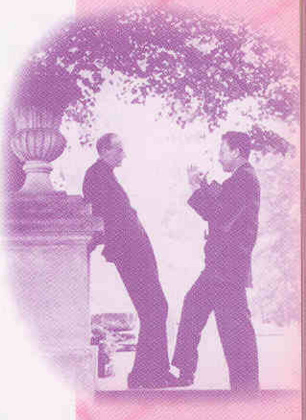
Richard Feynman (1918–88), the greatest physicist of his generation, was one of the founders of quantum electrodynamics. He invented the diagrams that describe the quantum world, and for 40 years he inspired young physicists with his teaching. He received the Nobel Prize in 1965 and was a key member of the panel that investigated the 1986 *Challenger* disaster.





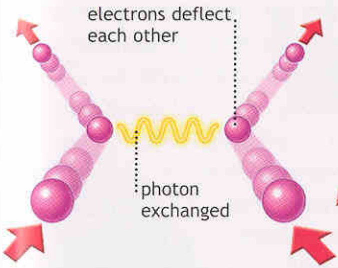
getting a kick

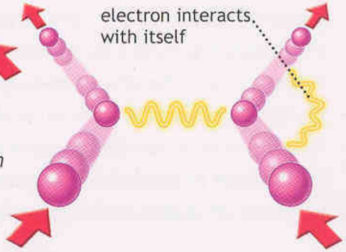
"World lines" on a Feynman diagram represent the positions of particles in space over time. A moving electron can emit or absorb a photon, causing it to change direction.



theoreticians

Seen here with Richard Feynman (right), Paul Dirac (1902–84) developed an early version of QED, and came up with the idea of antiparticles. Feynman conceived a complete theory of QED, finishing the job started by Dirac.





electron interactions

Two electrons can exchange a photon, so that they repel each other. An electron can also interact with itself.